



CAPSTONE PROJECT — HIGH-LEVEL OVERVIEW

AI Assistant Integration IISc Alumni Store

Conversational commerce, demonstrated end-to-end on a working storefront

Prepared for project review and stakeholder
presentation

Purpose of This Report

This report summarizes how a conversational AI assistant was integrated into an e-commerce storefront for this capstone. The **IISc Alumni Store** — a full working storefront with catalog, cart, checkout, and order history — serves as a realistic, dummy e-commerce platform to demonstrate the integration, rather than being the subject of the capstone itself. The focus is the pattern: how a conversational assistant is added to an existing storefront, what it can do, and how a customer's message flows through the system to become a reply.

This is a high-level summary for project review and presentation. It intentionally omits implementation detail, environment configuration, and hardening work that sit outside this phase.

Why This Capstone

Customers increasingly expect to self-serve on e-commerce sites — track an order, ask about a product, or get help — without leaving the page they are on or waiting for a human agent. This capstone demonstrates that experience end-to-end: a storefront with a conversational assistant integrated as a first-class feature, backed by a coordinated system of specialist AI agents rather than a single scripted chatbot.

Solution at a Glance

- The storefront is a standard e-commerce web application — browsing, cart, checkout, order history, and profile.
- A floating assistant is reachable from every page. Opening it does not interrupt shopping — the store stays visible and usable underneath.
- Behind one integration point, the assistant is powered by a coordinated system of specialist AI agents, each responsible for one category of customer need.

Integration Architecture & Flow

The diagram below shows how a customer's message travels from the storefront to the AI system and back, at a level suitable for a stakeholder walkthrough.

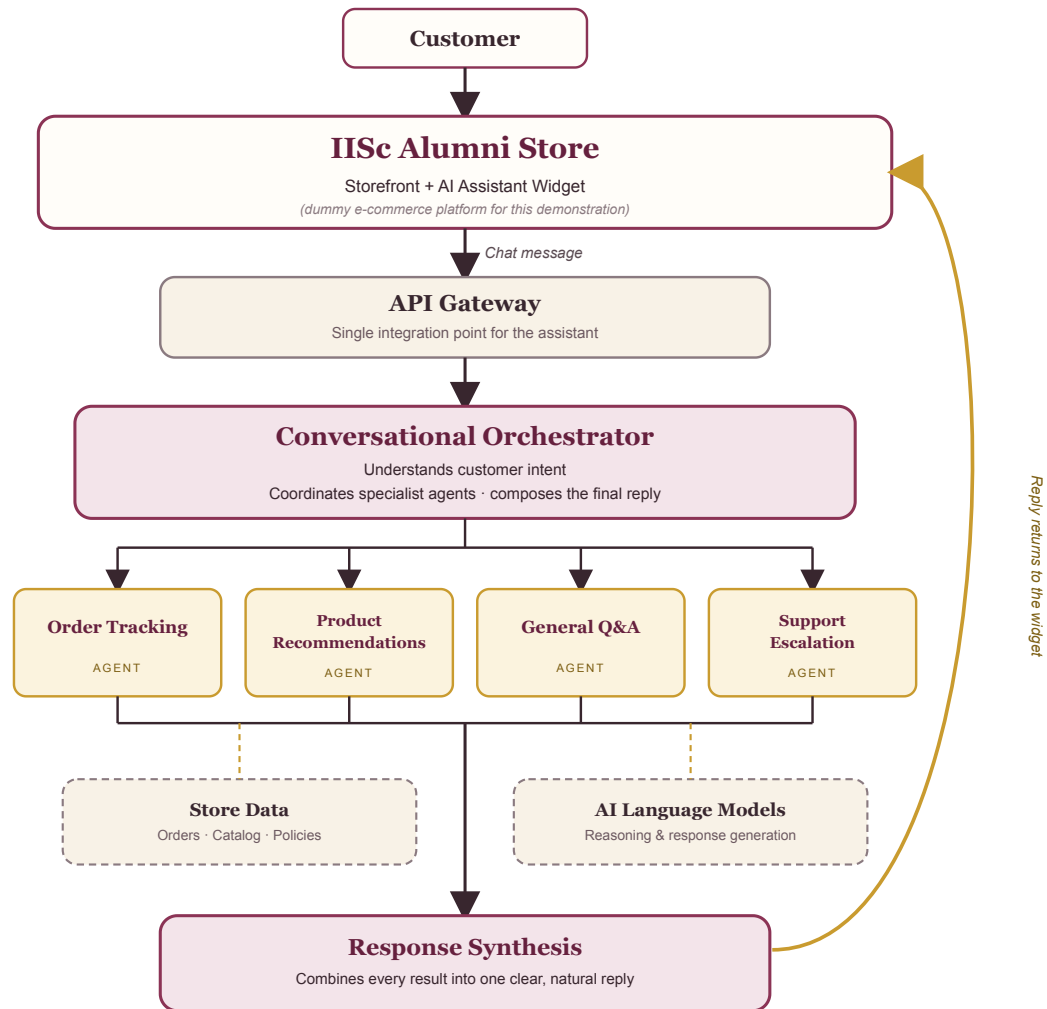


Figure 1 — End-to-end flow of a customer message through the integrated assistant

How a Conversation Flows

- 1 The customer opens the assistant from anywhere in the store and asks a question in plain language.
- 2 The storefront sends the message to the AI backend through a single, well-defined integration point — the storefront itself holds no built-in "intelligence"; it relays the conversation and displays the reply.
- 3 An orchestrator interprets what the customer is asking — including compound requests that combine more than one need, such as an order question and a gift suggestion in the same message.

- 4 Each part of the request is handed to the right specialist agent: order tracking, product recommendations, general product/policy questions, or escalation to human support.
- 5 Specialist agents draw on the store's own data (orders, catalog, policies) and on AI language models to work through the request.
- 6 The orchestrator combines every specialist's result into one coherent, on-brand reply.
- 7 The reply returns through the same integration point and appears in the assistant panel, continuing the conversation.

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Core Capabilities

Capability	What It Does
Order Tracking	Looks up an order by ID and explains its current status and expected delivery.
Product Recommendations	Suggests products by need, budget, or occasion, drawing on the live catalog.
Product & Policy Q&A	Answers general questions using the store's own catalog details and policy documents.
Support Escalation	Recognizes when a human is needed and raises a support ticket automatically.

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Customer Experience Design

- **Always available, never intrusive** — a floating launcher opens an inline panel; the store stays visible and usable underneath.
- **Guided start** — suggested prompts help a customer begin without having to type a first question.
- **Natural conversation** — replies appear progressively, with a clear indicator while the assistant is composing a response.
- **More than one way to talk to it** — customers can type or use voice input.
- **Continuity** — conversation history persists for the session, with a one-click option to start over.

Why a Multi-Agent Approach

- Each specialist agent focuses on one job, keeping answers grounded in real store data rather than open-ended guessing.
- Compound questions are automatically split and routed, so one message can engage multiple specialists at once and still come back as a single, coherent reply.
- Language-model usage is layered by task: lighter, faster models handle understanding and routing, while a more capable model is reserved for the reply the customer actually reads — balancing responsiveness with quality.
- Every request is traceable end-to-end, so the team can see exactly how a given reply was produced.

Scope of This Demonstration

This phase of the capstone focused on the integration architecture and customer experience described in this report. Environment configuration, production security hardening, and incremental response delivery are being addressed as separate, ongoing workstreams and are outside the scope of this report.

Outcome

This capstone demonstrates a realistic, extensible pattern for embedding a multi-agent conversational assistant into an existing e-commerce platform — reachable through a single integration point, with the storefront requiring no built-in intelligence of its own. The pattern generalizes to any storefront with minimal change to the store itself.